



Sustainable
Forests And
Livelihoods Project

The Government of the People's Republic of Bangladesh
Office of the Range Officer, Nolchira Range
Coastal Forest Division Noakhali

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PLANTATION JOURNAL

"Sustainable Forest & Livelihood (SUFAL) Project"

FY-2020-2021
250.0 Hect.

Habibiya Range
Coastal Forest Division, Noakhali.

PLANTATION JOURNAL

"Sustainable Forest & Livelihood (SUFAL) Project"

A	Basic Information												
1.	Name of the plantation with Year	Mangrove Plantation (2020-2021) (250.0 Hect)											
2.	Source of Fund	World Bank											
3.	Location	District: Noakhali	Union: N/A										
		Upazila: Hatia	Char/Village/Mouja: New Char Nur Islam										
4.	Name & Designation of the Offices in Charge (RO & BO)	Md. Nahid Hasan Ranger Officer Habibiya Range, Coastal Forest Division, Noakhali.	Mr. Giyash Uddin, Fr Beat Officer, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.										
5	Name & Designation of the Staff Involved in the Plantation Activities	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%;">1. Bipul Krishna Das Divisional Forest Officer Coastal Forest Division, Noakhali.</td> <td style="width: 50%;">2. Kazi Ttarikur Rahman Assistant Conservator of Forest Coastal Forest Division, Noakhali.</td> </tr> <tr> <td>1. Md. Nahid Hasan Ranger Officer, Habibiya Range Coastal Forest Division, Noakhali.</td> <td>2. Mr. Giyash Uddin, Fr Beat Officer, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.</td> </tr> <tr> <td>Mr. Himangso Kumar Roy Forest Guard, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.</td> <td>Md. Belayet Hossen Boat Man, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.</td> </tr> <tr> <td>Md. Abdul Khalek Plantation Mali, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.</td> <td>Md. Jahangir Hosen Plantation Mali, Habibiya Range, Coastal Forest Division, Noakhali.</td> </tr> <tr> <td>Md. Salamot Ullah Laskar Habibiya Range, Coastal Forest Division, Noakhali.</td> <td>Md. Abul Kalam Azad Speed Boat Driver Habibiya Range, Coastal Forest Division, Noakhali.</td> </tr> </table>		1. Bipul Krishna Das Divisional Forest Officer Coastal Forest Division, Noakhali.	2. Kazi Ttarikur Rahman Assistant Conservator of Forest Coastal Forest Division, Noakhali.	1. Md. Nahid Hasan Ranger Officer, Habibiya Range Coastal Forest Division, Noakhali.	2. Mr. Giyash Uddin, Fr Beat Officer, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.	Mr. Himangso Kumar Roy Forest Guard, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.	Md. Belayet Hossen Boat Man, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.	Md. Abdul Khalek Plantation Mali, Bashkhali Beat Habibiya Range, Coastal Forest Division, Noakhali.	Md. Jahangir Hosen Plantation Mali, Habibiya Range, Coastal Forest Division, Noakhali.	Md. Salamot Ullah Laskar Habibiya Range, Coastal Forest Division, Noakhali.	Md. Abul Kalam Azad Speed Boat Driver Habibiya Range, Coastal Forest Division, Noakhali.
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B	Characteristic of the Plantation Site												
1.	Area of the Plantation	250.0 ha Mangrove Plantation at New Char Nur Islam in the year of 2020-2021											
2	Edaphic Condition	Physiographically the Char is part of the young Meghna Estuarine Flood Plain. Hobgoblin, the site is soft and clayey and it is also comparatively compact and harder as well as at the same time subjected to more saline condition. These soils permit easy penetration of mangrove tree roots because of being quite fertile.											

3	Climate Condition	The New Char Nur Islam has a tropical monsoon climate, characterized by basically four seasons per year. The average temperature usually ranges between 11 ° to 29 ° C in winter and 21 ° to 38 ° C during summer. The average annual rainfall is 3257 mm and varies from 2500 mm to 4500 mm. Cyclones pass regularly, particularly at the beginning and at the end of the monsoon. The Char is slightly raised flat land which generally inundates twice a day noticeably from March to December. The rate of silt deposition is higher particularly from November to February roughly 3"-4".
4	Brief History: Plantation Techniques	The area was surveyed and demarcated by cutting Urigrass and some wooden post were firmly established along with the boundary line. Then the bar rooted Keora seedlings were collected from nursery. The seedling were planted by making Thales in line after cutting Urigrass while hole had been made with spacing 1.5m×1.5m where necessary. In addition, the roots of the seedlings were inserted upto the collar region into the soil and pressed hard from all sides so that no empty spaces were left.
5	Selection Criteria of the Planting Site	The plantation site has been selected through the following criteria. Land suitability: The New Char Nur Islam is newly accreted mud flats (char land). This site is inundated at each high tide, twice per day. This site has lot of fine mud on the surface that makes walking difficult with up to knee deep mud. This site starts bearing <i>Dhanshi</i> grasses. This area that has a thicker clayey layer and lower salinity are most suitable for planting with <i>Keora</i> (<i>Sonneratia apetala</i>). One should be able to walk on this site with less difficulty than the potential Keora planting sites. The purpose of Mangrove plantations is for protection to improve the resilience of coastal communities and their livelihoods. The Mangrove plantations can't be felled and are retained for protection from tidal swell and cyclonic winds. Grazing Problem: It is known to that there entire are a number of chars in Noakhali and accreting new lands. These chars are threat to the grazing problems. But this is free from such unavoidable phenomenon. That is why this site is selected for the mangrove plantation. Biodiversity Conservation: The Coastal afforestation program will bring a result in the establishment of plantation rich in biodiversity and higher density of wildlife habitats. Natural regeneration of forest tree species with their associates will be enhanced. In addition, through the project area the condition of coastal forest in Bangladesh will extent. Sectoral Target: The people residing in the project area are exposed to frequent cyclones and tidal surge which makes them highly vulnerable to recent climate change effects. As a result, their socio-economic condition is less secured. Beside this, the land mass developed due to accretion of the sediments discharged through Meghna estuary needs to be stabilized. On the other hand, the objectives of the forestry sector is to increase the total forest cover of the country and to optimize sustainable production of goods and services from the existing limited forest resources. The coastal area of the proposed project is such type of land where forestry resources could be developed through establishment of plantations. National Strategy: The proposed project is also aligned with the National Strategy; the <i>Bangladesh Sixth Five Year Plan (FY2018-FY2023)</i>. According to the plan, existing scattered coastal areas will be planted to increase productivity by promoting scientific management principles. In order to prevent

		the extent of damage by cyclones and tidal surges, the plan envisages creating a Coastal Green Belt and enhancing access to input materials, including seeds and seedlings. In addition, past climate disasters have shown that resilience to cyclones and storm surges is greater where mangrove greenbelts and buffer ecosystems exist. It is evident that mangroves can mitigate or reduce risk of natural disasters such as cyclones and tsunamis. a. Global Aspect: According to the IPCC report of 2013, approximately 95% human beings are the responsible for the climate change and Bangladesh is the home to the world's largest delta systems. Two third of the country is less than five meters above the sea level. It is vulnerable to short term climate shocks such as cyclone (DFID-2011) while the country is among top ten countries which are to climate change (Maple Craft-2013). As a result, human beings and natural resources are threats to climate change. So, there is no alternative without afforestation and reforestation in the coastal zones. That is why New Char Nur Islam is selected for raising mangrove plantation.
6	Climate Resilient Aspect of the Plantation	a. The Coastal forest, breakwater, seawall is to absorb the impact forces and to retard the flow of large storm waves and cyclone. A seawall, if tall enough, reflects the waves back out to sea. On the other hand, permeable structures, like breakwaters and coastal forests, partly reflect and partly transmit the water. In the case of a coastal forest, energy is progressively absorbed as it passes through the forest. Without the forest barrier, the tsunami/cyclone will run-up to a maximum height determined by the magnitude and nature of the seismic event that created the tsunami and local factors such as the coastal profile, offshore bathymetry and beach slope that modify the wave's force. Once the tsunami comes on shore, the amount of reduction in water depth, velocity, and force depends on how much water is reflected and energy is absorbed by the coastal forest. b. Through the afforestation in the project area biodiversity conservation, human security and so on will be ensured in the project area. Therefore, greenbelt of the project area by rising said plantation has been planned to bring back the balance of eco-system into the same char lands and to develop a friendly environment by raising coastal afforestation. It is for the fact that the land is suitable for mangrove and non-mangrove plantations where bio-physical and socio-economic conditions will improve. The New Char Nur Islam will be brought to work as a barrier during natural disasters by dint of the mangrove plantation.
C	Other Activities	
1	Delineation of the Planting Area	The New Char Nur Islam is a newly formed char land which has raised rapidly. The char is adjacent to the East-North side of Char Nur Islam Keora plantation. The land initially devoid any vegetation except dens Urigrass as first colonizer. The soils of the site permit easy penetration of mangrove tree roots because of being fertile. The project site would be used for different types of species diversity plantation. If the coastal forest lands of a target area are properly replenished with greens, the existing wildlife habitats will increase and will be a nice environment for biodiversity conservation. This type of coastal forest lands can fight back the prevailing environmental threats and mitigate the negative impacts of the climate changes took place in these coastal zones. Thus, this type of coastal forest lands not only save the life and resources from natural disasters but also play important role in economic development.

3	GPS Reading of all Corners of the Planting Site	22°25'22.8" 91°20'22.2" 22°25'25.9" 91°20'39.2" 22°26'37.5" 91°22'20.0"	22°26'36.5" 91°21'02.8" 22°27'13.9" 91°21'35.5" 22°27'13.0" 91°22'18.5"	22°27'45.0" 91°22'33.0" 22°26'37.5" 91°22'33.0" 22°26'09.4" 91°21'06.5"	The GPS has been taken from six corners of the plantation site in presence of the RO BO, BM, PM as well as all other staff officer of the coastal Forest Division Noakhali.
4	Post-Planting Survey, Surveyor, Survey Technique & Record	Md. Shafiqul Islam Surveyor, FD-Noakhali. The Survey has been completed through the technique of GPS reading and it is firstly recorded in the note book after that the record is shown in the Map.			

Officer's Signatures

Name
Mr. Giyash Uddin, Fr

Designation
Beat Officer, Bashkhali Beat
Habibiya Range,
Coastal Forest Division, Noakhali.

Signature

গিয়াস উদ্দিন
বিট কর্মকর্তা
বাশখালি বিট, হাবিবিয়া রেঞ্জ
উপকূলীয় বন বিভাগ, নোয়াখালী।

Mr. Md. Nahid Hasan,

Ranger Office,
Habibiya Range,
Coastal Forest Division, Noakhali.

Md. Nahid Hasan
Range Officer, Habibiya Range
Coastal Forest Division, Noakhali

Mr. Kazi Tariqur Rahman,
ACF

Assistant Conservator of Forest
Noakhali,
Coastal Forest Division, Noakhali

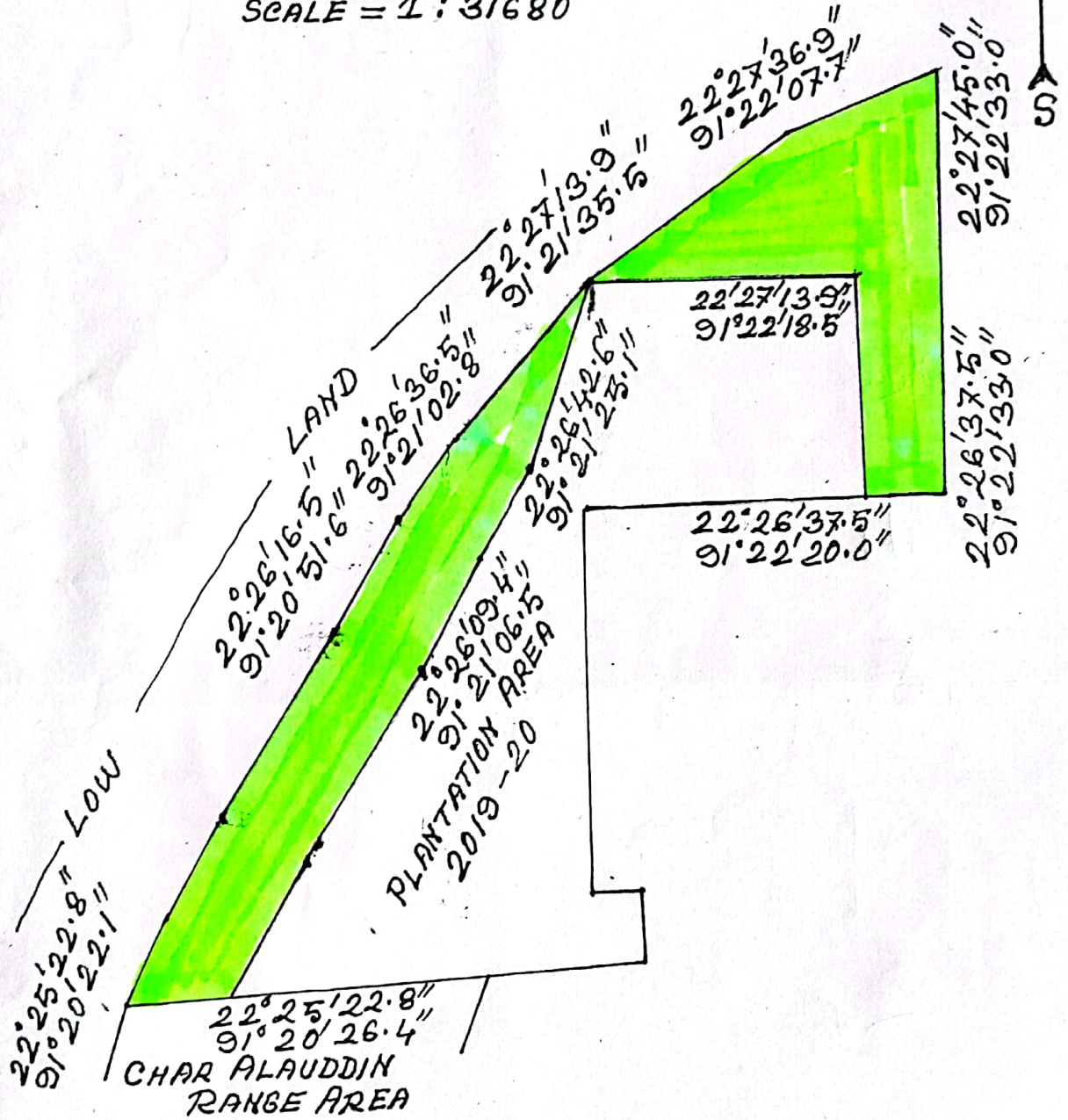
কাজী তারিকুর রহমান
সহকারী বন সংরক্ষক (নোয়াখালী জেলা)
উপকূলীয় বন বিভাগ, নোয়াখালী

Mr. Bipul Krishna Das, DCF

Divisional Forest Officer, (DFO)
Coastal Forest Division, Noakhali

(Bipul Krishna Das)
Divisional Forest Officer
Coastal Forest Division,
Noakhali.

POST PLANTING MAP OF MANGROVE PLANTATION AREA
 BEAT: BASHKHALI, RANGE: HABIBIYA
 COASTAL FOREST DIVISION NOAKHALI
 NAME OF CHAR: NEW CHAR NURISLAM
 YEAR: 2020-21
 AREA: 250.0 HEC.
 SCALE = 1 : 31680



REF:~

1	POST PLANTING AREA - 2020-21	
2	RIVER	
3	KHAL	
4		

মোঃ শাফিকুল ইসলাম
 সার্ভেয়ার
 উপকূলীয় বন বিভাগ নোয়াখালী।

Md. Nahid Hasan
 Range Officer, Habibiya Range
 Coastal Forest Division, Noakhali

সিরাস উদ্দিন
 বিট কর্মকর্তা
 বাশখালি বিট, হাবিবিয়া রেঞ্জ
 উপকূলীয় বন বিভাগ, নোয়াখালী।

Inspection Sheet

Date	Name & Designation	Remarks
২৫/০২/২০	কাজী তারিকুর রহমান ACF	অন্য নিম্ন আয়তনবর্তী এল জিওর্জি, হাবিসিয়া ব্লক ও যজ্ঞাও মফসসহ ২০২০-২১ সালে "সুফল" প্রকল্পের আওতাও হাবিসিয়া ব্লকের বাগানখানা বিলি সুদিত ২৫০.০ টি মাসেরো বাগান পরিচালনা করি। বাগানের বর্তমান অবস্থা সন্তোষজনক। সুফলসহান সুফল কাষ চাষা রয়েছে। সুফল বেলগিত জীবিত ১৫০০ পত্রমান প্রায় ৭০%। তবে সুফলসহান সুফল মাটিমে তা ক্ষত তাজে প্রচলিত কল-হুহু। জামাফল মাথো কথা বানি এল তাদেরে সতর্ক চাষ বানি। বাগানটি যাতে দক্ষ বাগান পরিচালনা সে ব্যাপারে ব্লক জিওর্জি, হাবিসিয়া-ব্লকে সহায়তা দৃষ্টি রাখাও বলা হয়েছে।  ২৫/০২/২০ কাজী তারিকুর রহমান সহকারী বন সংরক্ষক (নোয়াখালী জেলা) উপকূলীয় বন বিভাগ, নোয়াখালী

Plantation Picture



Abon
Md. Nahid Hase
Range Officer, Habibiya
Coastal Forest Division, 2



জিয়াস উদ্দিন
জিএসও
বিশ্বজলি বি. হাবিবিয়া ও
উদ্ভিদ বন বিভাগ, নোয়া